Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - II
Course Code : CSE 0533 1201 (CSE) & PHY 0533 1201 (EEE)

Term: Spring – 2024	Date: 10.01.2024	Time: 03.30 PM	Room: 327	Activity: Lecture 4

Topics to be discussed: Kinetic theory of gases & Its applications
Mean free path, Van der Waal's equation of state, Review of the First law of thermodynamics

Target CLO & PLO : CLO2-3 & PLO2-5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define elastic collision, free path & mean free path
- ❖ State and explain Van der Waal's equation of state
- ❖ Distinguish real & ideal gases.
- ❖ State the first law of thermodynamics
- ❖ Solve assigned recommended mathematical problems

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion on some natural phenomenon	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

1) What is the unit of the mean free path?
   a) N b) m c) F d) Nm^{-1} e) mN^{-1}
2) How many corrections needed to derive the real gas equation from ideal gas equation?
   a) 2 b) 3 c) 4 d) 5 e) 6
3) Which part of the Van-der-Wall equation comes from intermolecular force?
   a) (V-b) b) RT c) nRT d) $P+(a^2/V)$ e) PV

Sample Questions (SAQ):

1. Define free path & mean free path.
2. Derive the equation for the Mean Free Path of gas molecules.
   Or, Show that, the Mean free path is inversely proportional to its density.
   Or, What are the factors on which mean free depends?
3. The diameter of the molecule of a gas is 3×10^{-10}m and number of molecules per cm^3 is 6×10^{20}. Calculate the mean free path of the molecules.
4. The mean free path of the molecules of a gas is $6 \times 10^{-8}\mathrm{m}$ and the diameter of a molecule is $2.5 \times 10^{-10}\mathrm{m}$. Find the number of molecules per m^3.
5. Distinguish the real and ideal gas. Is it possible to get an ideal gas in reality?
6. Why real gas does not obey the ideal characteristics?
   Or, Explain the Van-der Wall Equation for real gas.

Reference Books:
1. Giasuddin ahmed – Physics for Engineers (Part-1)
2. Halliday Resnick - Fundamental of Physics, 2nd edition

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 01-64
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 01-36